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Hiring Manager

Mercury Insurance

Dear Hiring Manager,

My name is Ajay Venkatesh. I'm finishing my M.S. in Computer Engineering at NYU, with production software experience at PwC in Bengaluru, a summer SDE internship at Amazon, and ongoing on-campus work at NYU's Novel AI Technologies. I'm writing about the MLOps Catastrophe Modeling and Geo-Analytics role.

The combination, productionizing ML models with high availability and reliability through automated pipelines, is exactly the work I've been doing. At Amazon I deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that autonomously reasons over operational logs to generate context-aware summaries, and implemented a **Model Context Protocol (MCP)** server that materially reduced query time and compute costs. I provisioned resilient **Infrastructure-as-Code** on **AWS CDK** across auto-scaling ECS clusters with multi-stage CI/CD and rollback controls, the same posture MLOps deployment demands.

On modeling and training: my Big Data project built a **PySpark / HDFS** pipeline over millions of flight records, training Random Forest and Gradient Boosted Tree models with rigorous EDA. I've also worked on **RoBERTa fine-tuning with custom LoRA** and **ResNet** architectures under strict parameter and memory budgets, and built RAG systems on **AWS Bedrock** with FAISS indexing.

On data pipelines and engineering: at NYU I engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with concurrency control, processing hundreds of thousands of records and powering **AWS QuickSight** dashboards. At PwC I parallelized batch ingestion on operational feeds that materially cut execution time. Reliable data plumbing feeding production models is the unglamorous part of MLOps that compounds.

Stack-wise: **Python, SQL, PyTorch, TensorFlow, MLflow**, plus **AWS** (Lambda, SageMaker, ECS Fargate, CDK, S3, IAM), **Docker, Kubernetes, Git**, and **GitHub** CI/CD workflows. Comfortable with cloud infrastructure for high-availability ML systems.

What pulls me toward Mercury is the operational seriousness. Catastrophe modeling for insurance runs on accuracy that translates into real-world risk pricing and claim outcomes, where careful MLOps engineering matters more than novelty.

I'd welcome the chance to discuss further. Thanks for considering my application.

Sincerely,

Ajay Venkatesh